Tutorial No : 8

Tutorial Title: Implement Binary search tree with operations Create, search, and recursive traversal.

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Program:

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
   int data;
   struct node *left;
   struct node *right;
};

struct node* createNode(int data)
{
   struct node *newNode;

   newNode = (struct node*)malloc(sizeof(struct node));
   newNode->data = data;
   newNode->left = NULL;
   newNode->right = NULL;

   return newNode;
}

struct node* insert(struct node *root, int data)
{
   if(root == NULL)
   {
      return createNode(data);
   }

   if(data < root->data)
   {
      root->left = insert(root->left, data);
   }
```

```
    else
    {
        root->right = insert(root->right, data);
    }

    return root;
}

void inorder(struct node *root)
{
    if(root != NULL)
    {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}

struct node* search(struct node *root, int key)
{
    if(root == NULL || root->data == key)
    {
        return root;
    }

    if(key < root->data)
    {
        return search(root->left, key);
    }
    else
    {
        return search(root->right, key);
    }
}

int main()
{
    struct node *root = NULL;
    struct node *result;
    int n, data, key, i;

    printf("Enter number of nodes: ");
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &data);
        root = insert(root, data);
    }

    printf("\nInorder Traversal: ");
    inorder(root);

    printf("\n\nEnter element to search: ");
    scanf("%d", &key);

    result = search(root, key);

    if(result != NULL)
    {
        printf("Element %d found in the tree.", key);
    }
    else
    {
        printf("Element %d not found in the tree.", key);
    }

    return 0;
}
```

Output:

```
Enter number of nodes: 5
Enter elements:
12 22 35 24 29

Inorder Traversal: 12 22 24 29 35

Enter element to search: 22
Element 22 found in the tree.
```